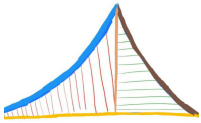
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE Class 12 Mathematics 2014

A. Formulae

1.1.1 Let $k \neq 0$. The binary operation $*$ on $\mathbb{R} - \{0\}$ defined by

$$a * b = \frac{ab}{k} \tag{1.1.1.1}$$

is associative, has k as its identity, and with $\phi(a) = \frac{a}{k}$,

$$\phi(a * b) = \phi(a)\,\phi(b) \tag{1.1.1.2}$$

Solution: From (1.1.1.1),

$$\phi(a * b) = \frac{1}{k} \cdot \frac{ab}{k} = \frac{a}{k} \cdot \frac{b}{k} \tag{1.1.1.3}$$

yielding (1.1.1.2). Since ϕ is one-to-one,

$$\phi((a * b) * c) = \phi(a)\,\phi(b)\,\phi(c) = \phi(a * (b * c)) \tag{1.1.1.4}$$

and $\phi(a * k) = \phi(a)\,\phi(k) = \phi(a)$, so k is the identity.

1.1.2 For $u \in [-1, 1]$,

$$\sin^{-1} u + \cos^{-1} u = \frac{\pi}{2} \tag{1.1.2.1}$$

Solution: Let $\theta = \sin^{-1} u \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Then $\frac{\pi}{2} - \theta \in [0, \pi]$ and

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta = u \implies \cos^{-1} u = \frac{\pi}{2} - \theta \tag{1.1.2.2}$$

yielding (1.1.2.1).

1.1.3 The principal value of $\cot^{-1}$ lies in $(0, \pi)$ and

$$\cot^{-1}(\cot\theta) = \theta, \quad \theta \in (0, \pi) \tag{1.1.3.1}$$

Similarly,

$$\tan^{-1}(\tan\theta) = \theta, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{1.1.3.2}$$

1.1.4 For any real x,

$$1 \pm \sin x = \left(\cos\frac{x}{2} \pm \sin\frac{x}{2}\right)^2 \tag{1.1.4.1}$$

Solution: Expanding the right hand side,

$$\cos^2\frac{x}{2} + \sin^2\frac{x}{2} \pm 2\sin\frac{x}{2}\cos\frac{x}{2} = 1 \pm \sin x \tag{1.1.4.2}$$

1.1.5 Let $\phi : S \to \mathbb{R}^m$ be any function. The relation

$$\mathbf{x} R \mathbf{y} \iff \phi(\mathbf{x}) = \phi(\mathbf{y}) \tag{1.1.5.1}$$

is an equivalence relation and the equivalence class of $\mathbf{x}$ is

$$[\mathbf{x}] = \{\mathbf{y} \in S : \phi(\mathbf{y}) = \phi(\mathbf{x})\} \tag{1.1.5.2}$$

Solution:

$$\phi(\mathbf{x}) = \phi(\mathbf{x}) \qquad \text{(reflexive)} \qquad (1.1.5.3)$$

$$\phi(\mathbf{x}) = \phi(\mathbf{y}) \implies \phi(\mathbf{y}) = \phi(\mathbf{x}) \qquad \text{(symmetric)} \qquad (1.1.5.4)$$

$$\phi(\mathbf{x}) = \phi(\mathbf{y}),\ \phi(\mathbf{y}) = \phi(\mathbf{z}) \implies \phi(\mathbf{x}) = \phi(\mathbf{z}) \qquad \text{(transitive)} \qquad (1.1.5.5)$$

1.1.6 For $n \times n$ matrices,

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \qquad (1.1.6.1)$$

Solution: Any invertible $\mathbf{A}$ is a product $\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_k$ of elementary matrices and, for a single elementary matrix, it is easy to verify that $\left|\mathbf{EB}\right| = \left|\mathbf{E}\right|\left|\mathbf{B}\right|$. Hence

$$\left|\mathbf{AB}\right| = \left|\mathbf{E}_1\right|\cdots\left|\mathbf{E}_k\right|\left|\mathbf{B}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \qquad (1.1.6.2)$$

upon substituting $\mathbf{B} = \mathbf{I}$. If $\mathbf{A}$ is singular, so is $\mathbf{AB}$ and both sides are 0.

1.1.7 Adding a multiple of one row of a matrix to another row does not change its determinant.

$$\left|\mathbf{EA}\right| = \left|\mathbf{A}\right|, \quad \mathbf{E} = \mathbf{I} + \lambda\mathbf{e}_i\mathbf{e}_j^\top,\ i \neq j \qquad (1.1.7.1)$$

Also, a common factor of a row can be taken out of the determinant, and the determinant can be expanded along any row.

Solution: $\mathbf{E}$ is triangular with unit diagonal, so $\left|\mathbf{E}\right| = 1$. Substituting in (1.1.6.1), we get (1.1.7.1).

1.1.8 The scalar triple product of three vectors is

$$[\mathbf{a},\mathbf{b},\mathbf{c}] = \mathbf{a}^\top(\mathbf{b}\times\mathbf{c}) = \begin{vmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{vmatrix} \qquad (1.1.8.1)$$

Solution: Expanding $\begin{vmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{vmatrix}$ along the first column, the cofactors of $\mathbf{a}$ are the components of $\mathbf{b}\times\mathbf{c}$.

1.1.9 The scalar projection of $\mathbf{a}$ on $\mathbf{b}$ is

$$\frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{b}\|} \qquad (1.1.9.1)$$

Also,

$$\|\mathbf{a}+\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + 2\mathbf{a}^\top\mathbf{b} + \|\mathbf{b}\|^2, \quad \cos\theta = \frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \qquad (1.1.9.2)$$

Solution: Write $\mathbf{a} = \lambda\mathbf{b} + \mathbf{a}_\perp$ with $\mathbf{b}^\top\mathbf{a}_\perp = 0$. Then

$$\lambda = \frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{b}\|^2} \implies \|\lambda\mathbf{b}\| = \frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{b}\|} \qquad (1.1.9.3)$$

The second result follows from expanding $(\mathbf{a}+\mathbf{b})^\top(\mathbf{a}+\mathbf{b})$.

1.1.10 A plane with normal vector $\mathbf{n}$ is

$$\mathbf{n}^\top\mathbf{x} = c \qquad (1.1.10.1)$$

A plane parallel to (1.1.10.1) and through $\mathbf{P}_0$ is $\mathbf{n}^\top\mathbf{x} = \mathbf{n}^\top\mathbf{P}_0$. The plane through

$\mathbf{A}, \mathbf{B}, \mathbf{C}$ has a normal vector $\mathbf{n}$ satisfying

$$\begin{pmatrix} (\mathbf{B}-\mathbf{A})^\top \\ (\mathbf{C}-\mathbf{A})^\top \end{pmatrix} \mathbf{n} = \mathbf{0}, \quad c = \mathbf{n}^\top \mathbf{A} \tag{1.1.10.2}$$

The distance of $\mathbf{P}$ from (1.1.10.1) is

$$d = \frac{\left|\mathbf{n}^\top \mathbf{P} - c\right|}{\|\mathbf{n}\|} \tag{1.1.10.3}$$

Solution: The foot of the perpendicular is $\mathbf{Q} = \mathbf{P} - \lambda \mathbf{n}$. Since $\mathbf{n}^\top \mathbf{Q} = c$,

$$\mathbf{n}^\top \mathbf{P} - \lambda \|\mathbf{n}\|^2 = c \implies \lambda = \frac{\mathbf{n}^\top \mathbf{P} - c}{\|\mathbf{n}\|^2} \tag{1.1.10.4}$$

yielding (1.1.10.3) from $d = \|\mathbf{P} - \mathbf{Q}\| = |\lambda| \, \|\mathbf{n}\|$. The normal vector in (1.1.10.2) is perpendicular to two directions in the plane.

1.1.11 The lines $\mathbf{x} = \mathbf{A}_1 + \lambda \mathbf{m}_1$ and $\mathbf{x} = \mathbf{A}_2 + \mu \mathbf{m}_2$ intersect if and only if

$$\mathbf{M} \begin{pmatrix} \lambda \\ \mu \end{pmatrix} = \mathbf{A}_2 - \mathbf{A}_1, \quad \mathbf{M} = \begin{pmatrix} \mathbf{m}_1 & -\mathbf{m}_2 \end{pmatrix} \tag{1.1.11.1}$$

is consistent, i.e.

$$\text{rank}\,(\mathbf{M}) = \text{rank}\left(\begin{pmatrix} \mathbf{M} & \mathbf{A}_2 - \mathbf{A}_1 \end{pmatrix}\right) \tag{1.1.11.2}$$

If $\text{rank}\,(\mathbf{M}) = 2$, the point of intersection is unique.

Solution: At a common point, $\mathbf{A}_1 + \lambda \mathbf{m}_1 = \mathbf{A}_2 + \mu \mathbf{m}_2$, which yields (1.1.11.1). This system has a solution if and only if $\mathbf{A}_2 - \mathbf{A}_1$ lies in the column space of $\mathbf{M}$, which is (1.1.11.2). If the columns are independent, the solution is unique.

1.1.12 If $\left|\mathbf{A}\right| \neq 0$, then $\mathbf{A}\mathbf{x} = \mathbf{b}$ has the unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}, \quad \mathbf{A}^{-1} = \frac{\text{adj}\,(\mathbf{A})}{\left|\mathbf{A}\right|} \tag{1.1.12.1}$$

Solution: Multiplying $\mathbf{A}\mathbf{x} = \mathbf{b}$ on the left by $\mathbf{A}^{-1}$ yields (1.1.12.1). The inverse exists since $\left|\mathbf{A}\right| \neq 0$.

1.1.13 The conditional probability of A given B is

$$\Pr(A|B) = \frac{\Pr(AB)}{\Pr(B)}, \quad \Pr(B) > 0 \tag{1.1.13.1}$$

If $E_1, E_2, \ldots, E_n$ are mutually exclusive and exhaustive events, then (Bayes' theorem)

$$\Pr(E_i|A) = \frac{\Pr(A|E_i)\Pr(E_i)}{\sum_{j=1}^{n} \Pr(A|E_j)\Pr(E_j)} \tag{1.1.13.2}$$

Solution: From (1.1.13.1), $\Pr(E_i A) = \Pr(A|E_i)\Pr(E_i)$. Since $A = \sum_j E_j A$ and

the E_jA are mutually exclusive, from (1.1.14.1),

$$\Pr(A) = \sum_{j=1}^{n} \Pr(A|E_j) \Pr(E_j) \tag{1.1.13.3}$$

Substituting in $\Pr(E_i|A) = \frac{\Pr(E_i A)}{\Pr(A)}$ yields (1.1.13.2).

1.1.14 For mutually exclusive events A and B,

$$\Pr(A + B) = \Pr(A) + \Pr(B) \tag{1.1.14.1}$$

1.1.15 For $y > 0$,

$$\frac{y'}{y} = (\ln y)', \quad \frac{y''}{y} - \left(\frac{y'}{y}\right)^2 = (\ln y)'' \tag{1.1.15.1}$$

Solution: Using the quotient rule,

$$\left(\frac{y'}{y}\right)' = \frac{y''y - (y')^2}{y^2} = \frac{y''}{y} - \left(\frac{y'}{y}\right)^2 \tag{1.1.15.2}$$

1.1.16 If $f'(x) > 0$ on an interval I, f is strictly increasing on I. If $f'(x) < 0$ on I, f is strictly decreasing on I. Also, if $f'(c) = 0$ and $f''(c) < 0$, f has a maximum at c.
Solution: For $a < b$ in I, by the mean value theorem there is a $\xi \in (a, b)$ such that

$$f(b) - f(a) = f'(\xi)(b - a) \tag{1.1.16.1}$$

whose sign is the sign of $f'(\xi)$.

1.1.17 The Laplace transform of $f(x)$, $x \geq 0$, is

$$F(s) = \mathcal{L}\{f(x)\} = \int_0^\infty f(x) e^{-sx}\, dx \tag{1.1.17.1}$$

It is linear, one-to-one, and

$$\mathcal{L}\{f'(x)\} = sF(s) - f(0) \tag{1.1.17.2}$$
$$\mathcal{L}\{xf(x)\} = -F'(s) \tag{1.1.17.3}$$

Solution: Integrating by parts,

$$\int_0^\infty f' e^{-sx}\, dx = \left[f e^{-sx}\right]_0^\infty + s\int_0^\infty f e^{-sx}\, dx = -f(0) + sF(s) \tag{1.1.17.4}$$

yielding (1.1.17.2). Differentiating (1.1.17.1) with respect to s,

$$F'(s) = \int_0^\infty (-x) f(x) e^{-sx}\, dx \tag{1.1.17.5}$$

which is (1.1.17.3).

1.1.18 Standard transforms, for $s > a$,

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{e^{ax}\} = \frac{1}{s-a}, \quad \mathcal{L}\{xe^{ax}\} = \frac{1}{(s-a)^2} \tag{1.1.18.1}$$

$$\mathcal{L}\{\ln x\} = -\frac{\gamma + \ln s}{s} \tag{1.1.18.2}$$

where γ is Euler's constant.

Solution: Since $\int_0^\infty e^{-(s-a)x}\,dx = \frac{1}{s-a}$, the first two follow. From (1.1.17.3),

$$\mathcal{L}\{xe^{ax}\} = -\frac{d}{ds}\left(\frac{1}{s-a}\right) = \frac{1}{(s-a)^2} \tag{1.1.18.3}$$

For $b > -1$, substituting $u = sx$,

$$\mathcal{L}\{x^b\} = \int_0^\infty x^b e^{-sx}\,dx = \frac{1}{s^{b+1}}\int_0^\infty u^b e^{-u}\,du = \frac{\Gamma(b+1)}{s^{b+1}} \tag{1.1.18.4}$$

Differentiating with respect to b at $b = 0$, using $\Gamma(1) = 1$ and $\Gamma'(1) = -\gamma$,

$$\mathcal{L}\{\ln x\} = \frac{\Gamma'(1) - \Gamma(1)\ln s}{s} \tag{1.1.18.5}$$

yielding (1.1.18.2).

1.1.19 For $|x| < 1$, the substitution $x = \tanh t$ gives

$$\frac{dx}{dt} = 1 - x^2, \quad t = \frac{1}{2}\ln\frac{1+x}{1-x} \tag{1.1.19.1}$$

Solution: Since $\frac{d}{dt}\tanh t = 1 - \tanh^2 t$, the first result follows. Solving $x = \frac{e^{2t}-1}{e^{2t}+1}$ for e^{2t} gives $e^{2t} = \frac{1+x}{1-x}$.

1.1.20 Standard integrals.

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \tag{1.1.20.1}$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C, \quad a \neq 0 \tag{1.1.20.2}$$

$$\int \sec^2 x\,dx = \tan x + C, \quad \int \operatorname{cosec}^2 x\,dx = -\cot x + C \tag{1.1.20.3}$$

Solution: For (1.1.20.2), substituting $x = a\tan\theta$, $dx = a\sec^2\theta\,d\theta$,

$$\int \frac{a\sec^2\theta}{a^2\left(1+\tan^2\theta\right)}\,d\theta = \frac{1}{a}\int d\theta = \frac{\theta}{a} + C \tag{1.1.20.4}$$

1.1.21 For any a, b,

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b) \tag{1.1.21.1}$$

Solution: Expanding $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ yields (1.1.21.1).

1.1.22 (King's property)

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx \tag{1.1.22.1}$$

Solution: Substituting $t = a+b-x$, $dx = -dt$, the limits $x = a, b$ become $t = b, a$.

$$\int_a^b f(x)\,dx = -\int_b^a f(t)\,dt = \int_a^b f(t)\,dt \tag{1.1.22.2}$$

1.1.23 In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$,

$$\iint_D dx\,dy = \iint_D r\,dr\,d\theta \tag{1.1.23.1}$$

Solution: The Jacobian determinant is

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r \tag{1.1.23.2}$$

1.1.24 (Corner point theorem) The maximum of $Z = \mathbf{c}^\top \mathbf{x}$ over a bounded polygon with vertices $\mathbf{v}_1, \ldots, \mathbf{v}_m$ is attained at a vertex.
Solution: Any point of the polygon is $\mathbf{x} = \sum_i \theta_i \mathbf{v}_i$, with $\theta_i \geq 0$, $\sum_i \theta_i = 1$. Then

$$\mathbf{c}^\top \mathbf{x} = \sum_i \theta_i \mathbf{c}^\top \mathbf{v}_i \leq \max_i \mathbf{c}^\top \mathbf{v}_i \tag{1.1.24.1}$$

1.1.25 For a right circular cone of slant height l, semi-vertical angle α, radius r and height h,

$$r = l\sin\alpha, \quad h = l\cos\alpha, \quad V = \frac{\pi}{3}r^2 h = \frac{\pi}{3}\left(l^2 - h^2\right)h \tag{1.1.25.1}$$

Solution: The axis, radius and slant edge form a right triangle with hypotenuse l, so $r^2 + h^2 = l^2$, from which the above follows.

B. Problems

1.2.1 Let $*$ be a binary operation on the set of all non-zero real numbers, given by $a*b = \frac{ab}{5}$ for all $a, b \in \mathbb{R} - \{0\}$. Find the value of x, given that $2*(x*5) = 10$.

Solution: From (1.1.1.1) with $k = 5$, $\phi(a) = \frac{a}{5}$ and

$$\phi(2*(x*5)) = \phi(2)\,\phi(x)\,\phi(5) = \frac{2}{5}\cdot\frac{x}{5}\cdot 1 = \frac{2x}{25} \tag{1.2.1.1}$$

from (1.1.1.2). Also, $\phi(10) = 2$. Equating,

$$\frac{2x}{25} = 2 \implies x = 25 \tag{1.2.1.2}$$

The following code verifies the solution.

codes/Binary_Operation.py

1.2.2 If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$, then find the value of x.

Solution: Since $\sin^{-1}\frac{1}{5} \in \left(0, \frac{\pi}{2}\right)$ and $\cos^{-1} x \in [0, \pi]$, the argument of sin lies in $\left[0, \frac{3\pi}{2}\right]$, where $\sin\theta = 1$ only for $\theta = \frac{\pi}{2}$. So

$$\sin^{-1}\frac{1}{5} + \cos^{-1} x = \frac{\pi}{2} \tag{1.2.2.1}$$

From (1.1.2.1), $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$. Hence

$$\sin^{-1}\frac{1}{5} = \sin^{-1} x \implies x = \frac{1}{5} \tag{1.2.2.2}$$

The following code computes $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1} x\right) = 1$ and generates Fig. 1.2.2.1. Fig. 1.2.2.1.

codes/Inverse_Cos.py

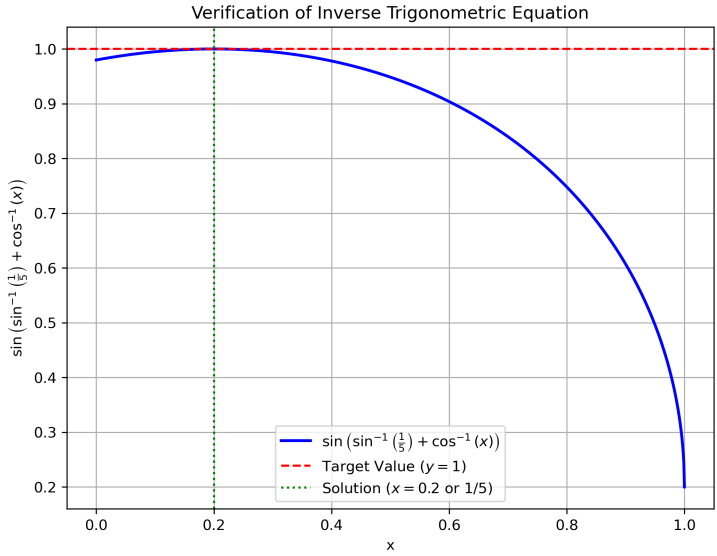


Fig. 1.2.2.1: Plot of $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1} x\right) = 1$

1.2.3 If $2\begin{pmatrix}3 & 4\\ 5 & x\end{pmatrix} + \begin{pmatrix}1 & y\\ 0 & 1\end{pmatrix} = \begin{pmatrix}7 & 0\\ 10 & 5\end{pmatrix}$, find $(x - y)$.

Solution: Equating the corresponding entries on both sides, the entries $7 = 7$ and $10 = 10$ hold identically, while

$$2x + 1 = 5, \quad 8 + y = 0 \implies \begin{pmatrix}2 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}4\\ -8\end{pmatrix} \tag{1.2.3.1}$$

yielding

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -8 \end{pmatrix} \implies x - y = 10 \tag{1.2.3.2}$$

The following code verifies the solution.

codes/Matrix_Equation.py

1.2.4 Solve the following matrix equation for x: $\begin{pmatrix} x & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix} = \mathbf{O}$.

Solution: Multiplying the row vector with the matrix,

$$\begin{pmatrix} x & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} x-2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix} \tag{1.2.4.1}$$

$$\implies x = 2 \tag{1.2.4.2}$$

The following code verifies the solution.

codes/Matrix_Equation_2.py

1.2.5 If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, write the value of x.

Solution: Evaluating both determinants,

$$2x^2 - 40 = 18 + 14 = 32 \implies x^2 = 36 \implies x = \pm 6 \tag{1.2.5.1}$$

The following code verifies the solution.

codes/Determinant_Equation.py

1.2.6 Write the antiderivative of $\left(3\sqrt{x} + \frac{1}{\sqrt{x}}\right)$.

Solution: From the standard integral of x^n in (1.1.20.2),

$$\int \left(3x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right) dx = 3 \cdot \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C = 2\sqrt{x}\,(x+1) + C \tag{1.2.6.1}$$

1.2.7 Evaluate : $\int \frac{dx}{9 + x^2}$

Solution: From (1.1.20.2) with $a = 3$,

$$\int \frac{dx}{9 + x^2} = \frac{1}{3} \tan^{-1} \frac{x}{3} + C \tag{1.2.7.1}$$

The following code verifies the solution.

codes/Arctan_Integral.py

1.2.8 Find the projection of the vector $\hat{i}+3\hat{j}+7\hat{k}$ on the vector $2\hat{i}-3\hat{j}+6\hat{k}$.

Solution: Let

$$\mathbf{a}=\begin{pmatrix}1\\3\\7\end{pmatrix},\quad \mathbf{b}=\begin{pmatrix}2\\-3\\6\end{pmatrix} \tag{1.2.8.1}$$

Then $\mathbf{a}^\top\mathbf{b}=2-9+42=35$ and $\|\mathbf{b}\|=\sqrt{4+9+36}=7$. From (1.1.9.1), the projection is

$$\frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{b}\|}=\frac{35}{7}=5 \tag{1.2.8.2}$$

The following code verifies the solution.

codes/Projection.py

1.2.9 If $\mathbf{a}$ and $\mathbf{b}$ are two unit vectors such that $\mathbf{a}+\mathbf{b}$ is also a unit vector, then find the angle between $\mathbf{a}$ and $\mathbf{b}$.

Solution: Given $\|\mathbf{a}\|=\|\mathbf{b}\|=\|\mathbf{a}+\mathbf{b}\|=1$, from (1.1.9.2),

$$1=1+2\mathbf{a}^\top\mathbf{b}+1 \implies \mathbf{a}^\top\mathbf{b}=-\frac{1}{2} \tag{1.2.9.1}$$

Then

$$\cos\theta=\frac{\mathbf{a}^\top\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|}=-\frac{1}{2} \implies \theta=\frac{2\pi}{3} \tag{1.2.9.2}$$

The following code verifies the solution.

codes/Vectors_Angle.py

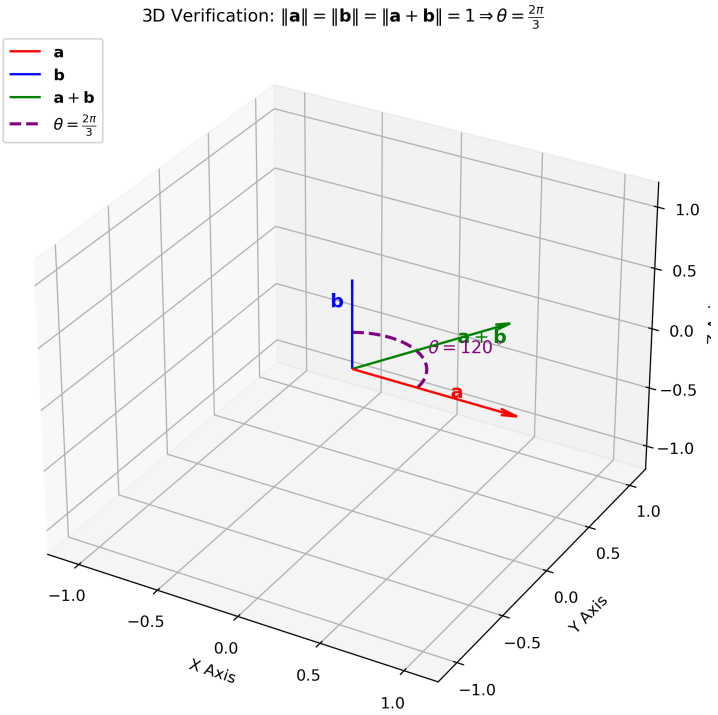


Fig. 1.2.9.1: Plot of $\frac{du}{dv}$

1.2.10 Write the vector equation of the plane, passing through the point (a, b, c) and parallel to the plane $\mathbf{r} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 2$.

Solution: The given plane is $\mathbf{n}^\top \mathbf{x} = 2$ in (1.1.10.1), with

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{1.2.10.1}$$

A parallel plane has the same normal vector. It passes through $\mathbf{P}_0 = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, so from (1.1.10.1),

$$\mathbf{n}^\top \mathbf{x} = \mathbf{n}^\top \mathbf{P}_0 \implies \mathbf{r} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = a + b + c \tag{1.2.10.2}$$

1.2.11 Let $A = \{1, 2, 3, \ldots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation. Also obtain the equivalence class $[(2, 5)]$.

Solution: Let

$$\mathbf{x} = \begin{pmatrix} a \\ b \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} c \\ d \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{1.2.11.1}$$

Then $a + d = b + c$ is the same as $a - b = c - d$, or

$$\mathbf{w}^\top \mathbf{x} = \mathbf{w}^\top \mathbf{y} \tag{1.2.11.2}$$

This is (1.1.5.1) with $\phi(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$. Hence R is reflexive, symmetric and transitive, i.e., an equivalence relation. The equivalence class of $\mathbf{x} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ is the set of points of $A \times A$ with $\mathbf{w}^\top \mathbf{y} = -3$, i.e. $d = c + 3$:

$$[(2,5)] = \{(1,4), (2,5), (3,6), (4,7), (5,8), (6,9)\} \tag{1.2.11.3}$$

The following code verifies the solution.

codes/Equivalence_Relation.py

1.2.12 Prove that $\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right) = \frac{x}{2}; x \in \left(0, \frac{\pi}{4}\right)$.

Solution: Let $c = \cos\frac{x}{2}$ and $s = \sin\frac{x}{2}$. For $x \in \left(0, \frac{\pi}{4}\right)$, $\frac{x}{2} \in \left(0, \frac{\pi}{8}\right)$, so $c > s > 0$. From (1.1.4.1),

$$\sqrt{1+\sin x} = c + s, \quad \sqrt{1-\sin x} = c - s \tag{1.2.12.1}$$

The argument of $\cot^{-1}$ becomes

$$\frac{(c+s)+(c-s)}{(c+s)-(c-s)} = \frac{2c}{2s} = \cot\frac{x}{2} \tag{1.2.12.2}$$

Since $\frac{x}{2} \in (0, \pi)$, from (1.1.3.1),

$$\cot^{-1}\left(\cot\frac{x}{2}\right) = \frac{x}{2} \tag{1.2.12.3}$$

The following code verifies the solution.

codes/Inverse_Trig.py

1.2.13 Using properties of determinants, prove that

$$\begin{vmatrix} 2y & y-z-x & 2y \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix} = (x+y+z)^3 \tag{1.2.13.1}$$

Solution: Let $s = x + y + z$ and let Δ be the determinant on the left hand side. Adding rows 2 and 3 to row 1 ($R_1 \to R_1 + R_2 + R_3$), from (1.1.7.1),

$$\Delta = \begin{vmatrix} s & s & s \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix} = s \begin{vmatrix} 1 & 1 & 1 \\ 2z & 2z & z-x-y \\ x-y-z & 2x & 2x \end{vmatrix} \tag{1.2.13.2}$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$,

$$\Delta = s \begin{vmatrix} 1 & 0 & 0 \\ 2z & 0 & -s \\ x-y-z & s & s \end{vmatrix} \tag{1.2.13.3}$$

Expanding along the first row,

$$\Delta = s\left(0 \cdot s - (-s)(s)\right) = s \cdot s^2 = (x+y+z)^3 \tag{1.2.13.4}$$

The following code verifies the solution.

codes/Determinant_Identity.py

1.2.14 Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ with respect to $\cos^{-1}\left(2x\sqrt{1-x^2}\right)$, when $x \neq 0$.

Solution: Let

$$u = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right), \quad v = \cos^{-1}\left(2x\sqrt{1-x^2}\right) \tag{1.2.14.1}$$

Substituting $x = \cos\theta$, $\theta \in (0,\pi)$, $\theta \neq \frac{\pi}{2}$, we have $\sqrt{1-x^2} = \sin\theta$. Then

$$u = \tan^{-1}(\tan\theta), \quad v = \cos^{-1}(\sin 2\theta) = \cos^{-1}\left[\cos\left(\frac{\pi}{2} - 2\theta\right)\right] \tag{1.2.14.2}$$

From (1.1.3.1) and the principal value of $\cos^{-1}$, $[0,\pi]$,

$$\frac{du}{d\theta} = 1, \quad \frac{dv}{d\theta} = \begin{cases} -2 & \theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \pi\right) \\ 2 & \theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \end{cases} \tag{1.2.14.3}$$

By the chain rule,

$$\frac{du}{dv} = \frac{du/d\theta}{dv/d\theta} = \begin{cases} -\frac{1}{2} & \frac{1}{\sqrt{2}} < |x| < 1 \\ \frac{1}{2} & 0 < |x| < \frac{1}{\sqrt{2}} \end{cases} \tag{1.2.14.4}$$

The following code computes $\frac{du}{dv}$ symbolically from the definitions and generates Fig. 1.2.14.1.

codes/Derivative_Ratio.py

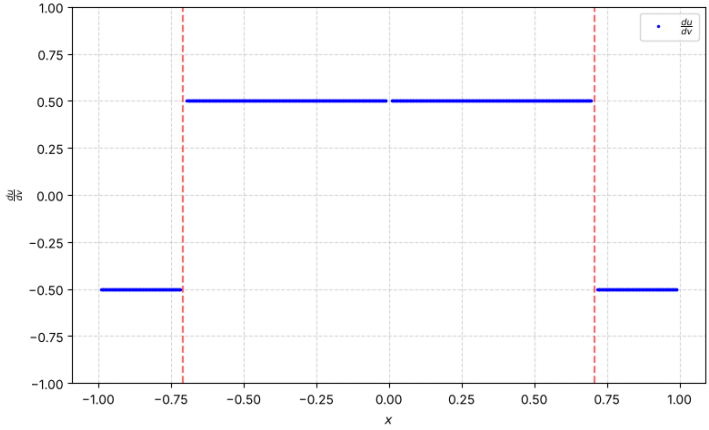


Fig. 1.2.14.1: Plot of $\frac{du}{dv}$

1.2.15 If $y = x^x$, prove that $\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$.

Solution: For $x > 0$, $y > 0$ and $\ln y = x \ln x$. Let $f = (\ln y)'$, so that

$$f = (x \ln x)' = 1 + \ln x \tag{1.2.15.1}$$

From (1.1.18.1) and (1.1.18.2),

$$F(s) = \mathcal{L}\{1 + \ln x\} = \frac{1}{s} - \frac{\gamma + \ln s}{s} \implies sF(s) = 1 - \gamma - \ln s \tag{1.2.15.2}$$

Let $g = xf = x + x \ln x$. Then $g(0) = 0$, $g' = f + xf'$ and, from (1.1.17.3), $G(s) = -F'(s)$. Using (1.1.17.2),

$$\mathcal{L}\{f + xf'\} = sG(s) - g(0) = -sF'(s) \tag{1.2.15.3}$$

yielding

$$\mathcal{L}\{xf'\} = -sF'(s) - F(s) = -(sF(s))' = \frac{1}{s} \tag{1.2.15.4}$$

Since $\frac{1}{s} = \mathcal{L}\{1\}$ and the transform is one-to-one, $xf' = 1$, i.e.

$$(\ln y)'' = f' = \frac{1}{x} \tag{1.2.15.5}$$

From (1.1.15.1),

$$\frac{y''}{y} - \left(\frac{y'}{y}\right)^2 = \frac{1}{x} \tag{1.2.15.6}$$

Multiplying by y,

$$y'' - \frac{(y')^2}{y} - \frac{y}{x} = 0 \tag{1.2.15.7}$$

The following code verifies the solution.

codes/Log_Derivative.py

1.2.16 Find the intervals in which the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is

a) strictly increasing
b) strictly decreasing

Solution: Differentiating,

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x-2)(x+1) \tag{1.2.16.1}$$

The critical points are $x = -1, 0, 2$. The sign of f' on each interval is given in the table below.

Interval	Sign of $f'(x)$	Nature of f
$(-\infty, -1)$	$-$	decreasing
$(-1, 0)$	$+$	increasing
$(0, 2)$	$-$	decreasing
$(2, \infty)$	$+$	increasing

TABLE 1.2.16: Sign of f'

From (1.1.16), f is

a) strictly increasing on $(-1, 0) \cup (2, \infty)$
b) strictly decreasing on $(-\infty, -1) \cup (0, 2)$

The following code verifies the solution and generates Fig. 1.2.16.1.

codes/Increasing_Function.py

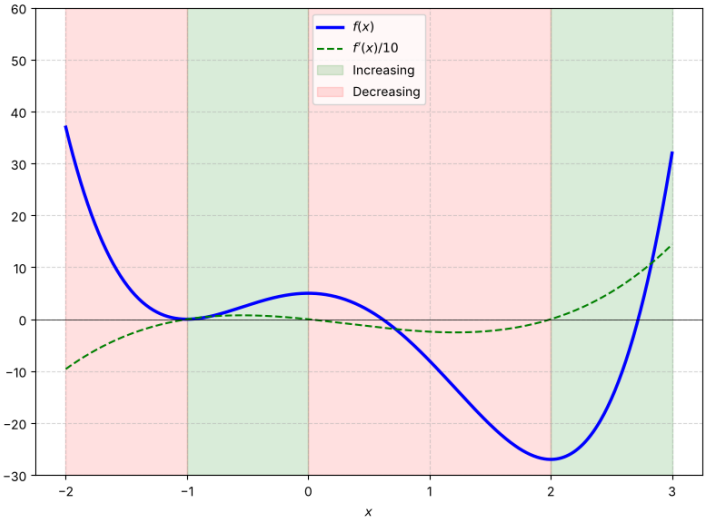


Fig. 1.2.16.1: Intervals of increase and decrease

1.2.17 Evaluate : $\int \dfrac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x}\, dx$

Solution: Let $a = \sin^2 x$, $b = \cos^2 x$, so that $a + b = 1$. From (1.1.21.1),

$$\sin^6 x + \cos^6 x = a^3 + b^3 = 1 - 3ab \tag{1.2.17.1}$$

Then the integrand is

$$\frac{1 - 3ab}{ab} = \frac{a + b}{ab} - 3 = \sec^2 x + \operatorname{cosec}^2 x - 3 \tag{1.2.17.2}$$

From the standard integrals,

$$\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x}\, dx = \tan x - \cot x - 3x + C \tag{1.2.17.3}$$

The following code verifies the solution.

codes/Trig_Integral.py

1.2.18 Find the particular solution of the differential equation $e^x \sqrt{1 - y^2}\, dx + \dfrac{y}{x}\, dy = 0$, given that $y = 1$ when $x = 0$.

Solution: Multiplying by $\frac{x}{\sqrt{1-y^2}}$, the variables get separated:

$$xe^x\, dx + \frac{y}{\sqrt{1 - y^2}}\, dy = 0 \tag{1.2.18.1}$$

Let $z = \sqrt{1-y^2}$. Then $z\,dz = -y\,dy$, and the equation becomes

$$\frac{dz}{dx} = xe^x, \quad z(0) = 0 \tag{1.2.18.2}$$

Taking the Laplace transform on both sides, from (1.1.17.2) and (1.1.18.1),

$$sZ(s) - z(0) = \frac{1}{(s-1)^2} \implies Z(s) = \frac{1}{s(s-1)^2} \tag{1.2.18.3}$$

Using partial fractions,

$$Z(s) = \frac{1}{s} - \frac{1}{s-1} + \frac{1}{(s-1)^2} \tag{1.2.18.4}$$

Taking the inverse transform from (1.1.18.1),

$$z = 1 - e^x + xe^x = (x-1)e^x + 1 \tag{1.2.18.5}$$

Hence the particular solution is

$$\sqrt{1-y^2} = (x-1)e^x + 1 \tag{1.2.18.6}$$

The following code finds the transform and verifies the solution.

codes/Separable_ODE.py

1.2.19 Solve the following differential equation

$$\left(x^2-1\right)\frac{dy}{dx} + 2xy = \frac{2}{x^2-1} \tag{1.2.19.1}$$

Solution: Since $\left(x^2-1\right)\frac{dy}{dx} + 2xy = \frac{d}{dx}\left[\left(x^2-1\right)y\right]$, let $w = \left(x^2-1\right)y$. The equation becomes

$$\frac{dw}{dx} = \frac{2}{x^2-1} \tag{1.2.19.2}$$

For $0 \le x < 1$, let $x = \tanh t$, $t \ge 0$. From (1.1.19.1),

$$\frac{dw}{dt} = \frac{dw}{dx}\cdot\frac{dx}{dt} = \frac{2}{x^2-1}\left(1-x^2\right) = -2, \quad w(0) = w_0 \tag{1.2.19.3}$$

Taking the Laplace transform on both sides, from (1.1.17.2),

$$sW(s) - w_0 = -\frac{2}{s} \implies W(s) = \frac{w_0}{s} - \frac{2}{s^2} \tag{1.2.19.4}$$

Taking the inverse transform, $w = w_0 - 2t$. Substituting $t = \frac{1}{2}\ln\frac{1+x}{1-x}$ from (1.1.19.1),

$$w = \ln\frac{1-x}{1+x} + w_0 \tag{1.2.19.5}$$

Hence, with $C = w_0$,

$$y = \frac{1}{x^2-1}\left[\ln\left|\frac{x-1}{x+1}\right| + C\right] \tag{1.2.19.6}$$

Differentiating, $\frac{d}{dx}\ln\left|\frac{x-1}{x+1}\right| = \frac{2}{x^2-1}$ for all $x \neq \pm 1$, so the same expression solves the equation on every interval. The following code finds the transform and verifies the solution.

codes/Linear_ODE.py

1.2.20 Prove that, for any three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$,

$$[\mathbf{a}+\mathbf{b}, \mathbf{b}+\mathbf{c}, \mathbf{c}+\mathbf{a}] = 2\,[\mathbf{a}, \mathbf{b}, \mathbf{c}] \tag{1.2.20.1}$$

Solution: Let $\mathbf{M} = \begin{pmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{pmatrix}$. Then

$$\begin{pmatrix}\mathbf{a}+\mathbf{b} & \mathbf{b}+\mathbf{c} & \mathbf{c}+\mathbf{a}\end{pmatrix} = \mathbf{MT}, \quad \mathbf{T} = \begin{pmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\end{pmatrix} \tag{1.2.20.2}$$

where the columns of $\mathbf{T}$ give the coefficients of $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Now,

$$\left|\mathbf{T}\right| = 1\,(1-0) - 0 + 1\,(1-0) = 2 \tag{1.2.20.3}$$

From (1.1.8.1) and (1.1.6.1),

$$[\mathbf{a}+\mathbf{b}, \mathbf{b}+\mathbf{c}, \mathbf{c}+\mathbf{a}] = \left|\mathbf{MT}\right| = \left|\mathbf{M}\right|\left|\mathbf{T}\right| = 2\,[\mathbf{a}, \mathbf{b}, \mathbf{c}] \tag{1.2.20.4}$$

The following code verifies the solution.

codes/Triple_Product.py

1.2.21 Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Also find their point of intersection.

Solution: The lines are $\mathbf{x} = \mathbf{A}_1 + \lambda\mathbf{m}_1$ and $\mathbf{x} = \mathbf{A}_2 + \mu\mathbf{m}_2$ with

$$\mathbf{A}_1 = \begin{pmatrix}-1\\-3\\-5\end{pmatrix}, \mathbf{m}_1 = \begin{pmatrix}3\\5\\7\end{pmatrix}, \mathbf{A}_2 = \begin{pmatrix}2\\4\\6\end{pmatrix}, \mathbf{m}_2 = \begin{pmatrix}1\\3\\5\end{pmatrix} \tag{1.2.21.1}$$

From (1.1.11.1), the augmented matrix is

$$\left(\begin{array}{cc|c}3 & -1 & 3\\ 5 & -3 & 7\\ 7 & -5 & 11\end{array}\right) \xrightarrow[R_3 \to 3R_3 - 7R_1]{R_2 \to 3R_2 - 5R_1} \left(\begin{array}{cc|c}3 & -1 & 3\\ 0 & -4 & 6\\ 0 & -8 & 12\end{array}\right) \xrightarrow{R_3 \to R_3 - 2R_2} \left(\begin{array}{cc|c}3 & -1 & 3\\ 0 & -4 & 6\\ 0 & 0 & 0\end{array}\right) \tag{1.2.21.2}$$

The ranks of the coefficient matrix and the augmented matrix are both 2, so (1.1.11.2) holds and the lines intersect. Back-substitution gives

$$\mu = -\frac{3}{2}, \quad 3\lambda = 3 + \mu \implies \lambda = \frac{1}{2} \tag{1.2.21.3}$$

The point of intersection is

$$\mathbf{A}_1 + \lambda \mathbf{m}_1 = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} \tag{1.2.21.4}$$

The following code verifies the solution and generates Fig. 1.2.21.1.

codes/Skew_Lines.py

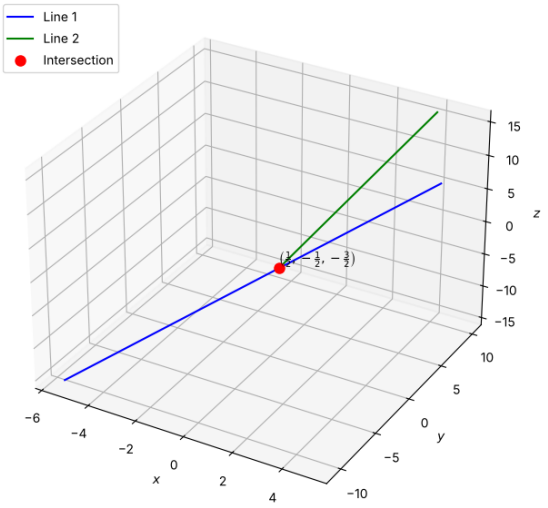


Fig. 1.2.21.1: Intersecting lines

1.2.22 Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls? Given that

a) the youngest is a girl.
b) atleast one is a girl.

Solution: Writing the outcomes as (elder, younger), the sample space is $\{GG, GB, BG, BB\}$ with probability $\frac{1}{4}$ each. Let

- A: both are girls $= \{GG\}$
- B: the youngest is a girl $= \{GG, BG\}$
- C: at least one is a girl $= \{GG, GB, BG\}$

Since $AB = AC = A$, from (1.1.13.1),

$$\Pr(A|B) = \frac{\Pr(A)}{\Pr(B)} = \frac{1/4}{2/4} = \frac{1}{2} \tag{1.2.22.1}$$

$$\Pr(A|C) = \frac{\Pr(A)}{\Pr(C)} = \frac{1/4}{3/4} = \frac{1}{3} \tag{1.2.22.2}$$

The following code simulates the result.

codes/Conditional_Probability.py

1.2.23 Two schools P and Q want to award their selected students on the values of Discipline, Politeness and Punctuality. The school P wants to award ₹ x each, ₹ y each and ₹ z each for the three respective values to its 3, 2 and 1 students with a total award money of ₹ 1,000. School Q wants to spend ₹ 1,500 to award its 4, 1 and 3 students on the respective values (by giving the same award money for the three values as before). If the total amount of awards for one prize on each value is ₹ 600, using matrices, find the award money for each value. Apart from the above three values, suggest one more value for awards.

Solution: The given conditions are

$$3x + 2y + z = 1000, \quad 4x + y + 3z = 1500, \quad x + y + z = 600 \tag{1.2.23.1}$$

or

$$\begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1000 \\ 1500 \\ 600 \end{pmatrix} \tag{1.2.23.2}$$

Let the coefficient matrix be $\mathbf{A}$. Then

$$\left|\mathbf{A}\right| = 3(1-3) - 2(4-3) + 1(4-1) = -5 \neq 0 \tag{1.2.23.3}$$

The adjoint of $\mathbf{A}$ gives

$$\mathbf{A}^{-1} = \frac{1}{5} \begin{pmatrix} 2 & 1 & -5 \\ 1 & -2 & 5 \\ -3 & 1 & 5 \end{pmatrix} \tag{1.2.23.4}$$

From (1.1.12.1),

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 & 1 & -5 \\ 1 & -2 & 5 \\ -3 & 1 & 5 \end{pmatrix} \begin{pmatrix} 1000 \\ 1500 \\ 600 \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix} \tag{1.2.23.5}$$

The awards are ₹ 100 for Discipline, ₹ 200 for Politeness and ₹ 300 for Punctuality. One more value that can be awarded is Honesty. The following code verifies the solution.

codes/Matrix_System.py

1.2.24 Show that the semi-vertical angle of the cone of the maximum volume and of given slant height is $\cos^{-1}\frac{1}{\sqrt{3}}$.

Solution: For a fixed slant height l, from (1.1.25.1), the volume is a function of the height h:

$$V(h) = \frac{\pi}{3}\left(l^2 h - h^3\right) \tag{1.2.24.1}$$

Differentiating,

$$V'(h) = \frac{\pi}{3}\left(l^2 - 3h^2\right) = 0 \implies h = \frac{l}{\sqrt{3}} \tag{1.2.24.2}$$

$$V''(h) = -2\pi h < 0 \tag{1.2.24.3}$$

So V is maximum at $h = \frac{l}{\sqrt{3}}$, from (1.1.16). Using $h = l\cos\alpha$,

$$\cos\alpha = \frac{h}{l} = \frac{1}{\sqrt{3}} \implies \alpha = \cos^{-1}\frac{1}{\sqrt{3}} \tag{1.2.24.4}$$

The maximum volume is $\frac{2\pi l^3}{9\sqrt{3}}$. The following code plots V as a function of α in Fig. 1.2.24.1.

codes/Cone_Volume.py

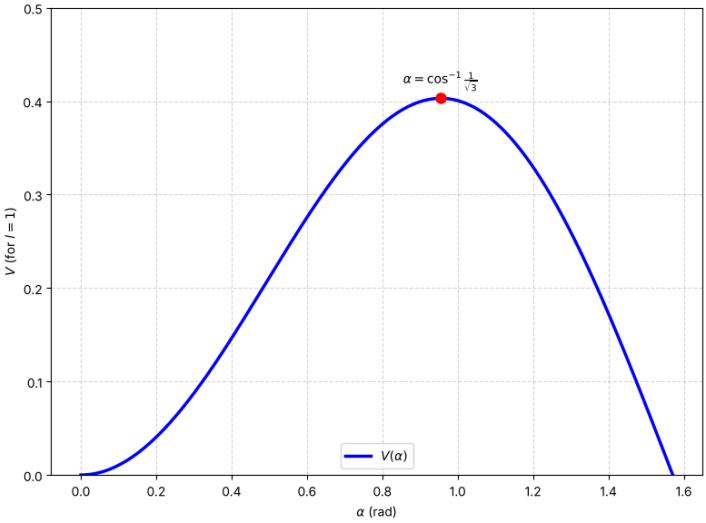


Fig. 1.2.24.1: Volume of the cone versus α

1.2.25 Evaluate : $\displaystyle\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$

Solution: Multiplying the numerator and denominator by $\sqrt{\sin x}$,

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\, dx \tag{1.2.25.1}$$

Here $a + b = \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$. From (1.1.22.1),

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}}\, dx \tag{1.2.25.2}$$

Adding the two expressions of I,

$$2I = \int_{\pi/6}^{\pi/3} dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} \implies I = \frac{\pi}{12} \tag{1.2.25.3}$$

The following code verifies the solution.

codes/Integral_Symmetry.py

1.2.26 Find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.

Solution: The line $y = x$ is $\theta = \frac{\pi}{4}$ and the circle is $r = \sqrt{32} = 4\sqrt{2}$. They meet at $(4, 4)$. The region is $0 \le r \le 4\sqrt{2}$, $0 \le \theta \le \frac{\pi}{4}$. From (1.1.23.1),

$$A = \int_0^{\pi/4} \int_0^{4\sqrt{2}} r\, dr\, d\theta = \frac{\pi}{4} \cdot \frac{32}{2} = 4\pi \tag{1.2.26.1}$$

The following code computes the area and generates Fig. 1.2.26.1.

codes/Area_Polar.py

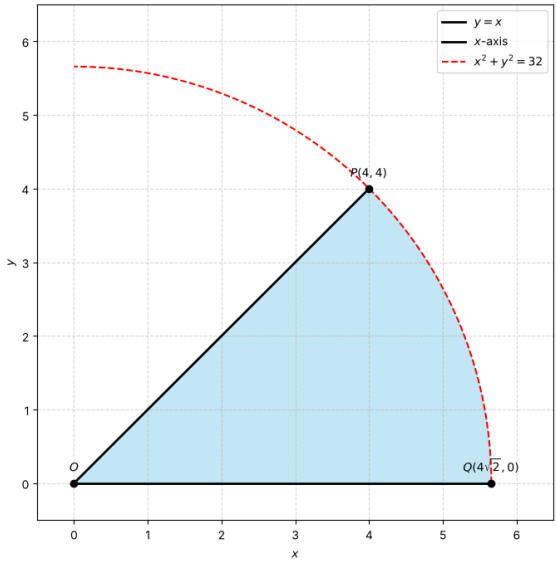


Fig. 1.2.26.1: Region enclosed by the x-axis, $y = x$ and the circle

1.2.27 Find the distance between the point $(7, 2, 4)$ and the plane determined by the points $A\,(2, 5, -3)$, $B\,(-2, -3, 5)$ and $C\,(5, 3, -3)$.

Solution: Let

$$\mathbf{A} = \begin{pmatrix} 2 \\ 5 \\ -3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -2 \\ -3 \\ 5 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 5 \\ 3 \\ -3 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 7 \\ 2 \\ 4 \end{pmatrix} \tag{1.2.27.1}$$

Then $\mathbf{B} - \mathbf{A} = \begin{pmatrix} -4 \\ -8 \\ 8 \end{pmatrix}$ and $\mathbf{C} - \mathbf{A} = \begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$. From (1.1.10.2),

$$\begin{pmatrix} -4 & -8 & 8 \\ 3 & -2 & 0 \end{pmatrix} \mathbf{n} = \mathbf{0} \xrightarrow[R_2 \to R_2 - 3R_1]{R_1 \to -\frac{R_1}{4}} \begin{pmatrix} 1 & 2 & -2 \\ 0 & -8 & 6 \end{pmatrix} \xrightarrow[R_1 \to R_1 + \frac{R_2}{4}]{R_2 \to -\frac{R_2}{8}} \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{3}{4} \end{pmatrix} \tag{1.2.27.2}$$

so $n_1 = \frac{n_3}{2}$, $n_2 = \frac{3n_3}{4}$ and we can take

$$\mathbf{n} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad c = \mathbf{n}^\top \mathbf{A} = 4 + 15 - 12 = 7 \tag{1.2.27.3}$$

The plane is $\begin{pmatrix} 2 & 3 & 4 \end{pmatrix} \mathbf{x} = 7$. From (1.1.10.3),

$$d = \frac{|14 + 6 + 16 - 7|}{\sqrt{4 + 9 + 16}} = \frac{29}{\sqrt{29}} = \sqrt{29} \tag{1.2.27.4}$$

The following code verifies the solution.

codes/Plane_Distance.py

1.2.28 A dealer in rural area wishes to purchase a number of sewing machines. He has only ₹ 5,760 to invest and has space for at most 20 items for storage. An electronic sewing machine cost him ₹ 360 and a manually operated sewing machine ₹ 240. He can sell an electronic sewing machine at a profit of ₹ 22 and a manually operated sewing machine at a profit of ₹ 18. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize his profit? Make it as a LPP and solve it graphically.

Solution: Let x electronic and y manually operated machines be purchased, $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. The problem is

$$\max Z = \begin{pmatrix} 22 & 18 \end{pmatrix} \mathbf{x} \quad \text{s.t.} \quad \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \mathbf{x} \le \begin{pmatrix} 48 \\ 20 \end{pmatrix}, \quad \mathbf{x} \ge \mathbf{0} \tag{1.2.28.1}$$

where the first constraint is $360x + 240y \le 5760$. The feasible region in Fig. 1.2.28.1 is a polygon. The vertex B where the two constraints are tight is

$$\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 48 \\ 20 \end{pmatrix} \implies \mathbf{x} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 48 \\ 20 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix} \tag{1.2.28.2}$$

The value of Z at the vertices is given in the table below.

Vertex	(x, y)	$Z = 22x + 18y$
O	$(0, 0)$	0
A	$(16, 0)$	352
B	$(8, 12)$	392
C	$(0, 20)$	360

TABLE 1.2.28: Profit at the vertices

From (1.1.24), the maximum profit is ₹ 392, when 8 electronic and 12 manually operated machines are purchased. The following code solves the LPP and generates Fig. 1.2.28.1.

codes/LPP.py

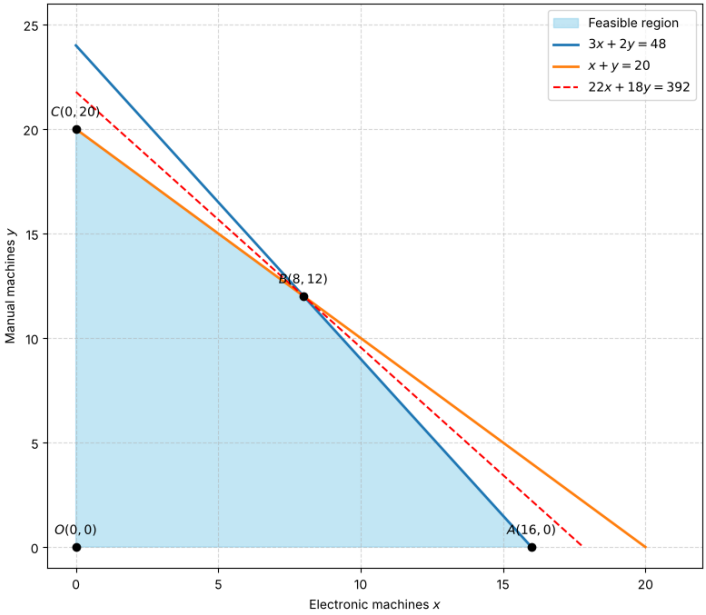


Fig. 1.2.28.1: Feasible region and the optimal vertex

1.2.29 A card from a pack of 52 playing cards is lost. From the remaining cards of the pack three cards are drawn at random (without replacement) and are found to be all spades. Find the probability of the lost card being a spade.

Solution: Let

- E_1: the lost card is a spade, $\Pr(E_1) = \frac{13}{52} = \frac{1}{4}$
- E_2: the lost card is not a spade, $\Pr(E_2) = \frac{3}{4}$
- A: three spades are drawn from the remaining 51 cards

If E_1 occurs, 12 spades remain, and if E_2 occurs, 13 spades remain. So

$$\Pr(A|E_1) = \frac{{}^{12}C_3}{{}^{51}C_3} = \frac{220}{{}^{51}C_3}, \quad \Pr(A|E_2) = \frac{{}^{13}C_3}{{}^{51}C_3} = \frac{286}{{}^{51}C_3} \tag{1.2.29.1}$$

From (1.1.13.2),

$$\Pr(E_1|A) = \frac{\frac{1}{4} \cdot 220}{\frac{1}{4} \cdot 220 + \frac{3}{4} \cdot 286} = \frac{220}{1078} = \frac{10}{49} \tag{1.2.29.2}$$

The following code verifies the solution.

codes/Bayes.py